

Dementia Prevalence Across Texas Counties and the Association with Minerals Found in Groundwater

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Abstract. Alzheimer's Disease and Related Dementias (ADRD) represent a growing public health burden in the United States, with prevalence rising substantially over the past two decades. Although lithium has long been used in psychiatric treatment, emerging evidence suggests that it may also reduce dementia risk. While lithium is not classified as an essential micronutrient, chronic low-dose exposure may have biologically meaningful effects on brain health. Prior epidemiologic, experimental, and mechanistic studies report inverse associations between lithium exposure and dementia outcomes, yet fine-scale U.S. analyses remain limited. This study integrates groundwater lithium concentration data with 2020 county-level dementia prevalence estimates to examine associations across Texas counties. County-level exposure metrics are analyzed using exploratory methods and multivariate regression with demographic covariates. Results evaluate whether higher lithium concentrations correspond to lower dementia prevalence, informing environmental health surveillance and population-level strategies for the prevention of neurodegenerative diseases.

1 Introduction

Alzheimer's Disease and Related Dementia (ADRD) represent a group of neurodegenerative conditions characterized by the progressive loss of cognitive function and memory. The profound effect of these diseases on patients, families, and healthcare systems is a rapidly growing public health burden. Alzheimer's Disease (AD) is the most common form, accounting for 60-80% of dementia cases (Alzheimer's Association, 2025). In the early 2000s, an estimated 3.4 million Americans were living with ADRD (Plassman et al., 2007). By 2025, this number has increased to approximately 7 million, with projections estimating nearly 13 million affected individuals by 2050 (Alzheimer's Association, 2025). While symptoms of dementia may be mild during early stages, they progressively worsen over time, with an average life expectancy of approximately five years following diagnosis. This rapid rise in prevalence, alongside escalating economic costs and caregiving burdens, underscores the urgency of identifying modifiable risk factors and preventative strategies for ADRD that may reduce the population-level disease burden.

Numerous studies have examined risk factors contributing to ADRD incidence and mortality, including age, lifestyle factors, genetics, and socioeconomic conditions. More recently, researchers have identified pronounced geographic disparities in ADRD outcomes, with certain U.S. states and regions experiencing disproportionately higher rates and mortality (Fletcher et al., 2024). These spatial patterns have prompted increased attention to environmental determinants of neurological health, including naturally occurring trace elements in drinking water. Drinking water represents a particularly important pathway for environmental exposure because it is ubiquitous, chronic, and largely unavoidable. Unlike dietary or behavioral risk factors, water-based exposures occur continuously over time and affect entire populations regardless of individual choice. Naturally occurring trace elements in groundwater vary substantially across geographic regions due to differences in geology, hydrology, and water sourcing practices, creating an opportunity to examine long-term exposure–outcome relationships.

Among these trace elements, lithium (Li), a naturally occurring alkali metal found in groundwater at low concentrations, has received growing attention for its possible neuroprotective effects. Lithium has long been used pharmacologically in psychiatric treatment. Early observations of reduced neurodegenerative outcomes among individuals receiving long-term lithium therapy motivated further investigation into lithium's potential neuroprotective effects beyond its established psychiatric use. Emerging research now suggests that low-level, naturally occurring lithium exposure through drinking water may be associated with reduced rates of dementia (Kessing et al., 2010). This distinction between therapeutic and environmental exposure is particularly important, as trace lithium levels in water are orders of magnitude lower than clinical doses yet may still have biologically meaningful effects over long periods of exposure. If true, this relationship has important implications for preventive public

health, environmental monitoring, and policy decisions related to drinking water standards.

Prior epidemiological studies in Europe and North America have reported inverse associations between lithium concentrations in drinking water and dementia outcomes. Mechanistic studies further suggest lithium's role in neuroprotection through pathways involving amyloid- β regulation, myelin integrity, and brain-derived neurotrophic factor (BDNF) (Motaghinejad et al., 2016; Depp et al., 2023). Additional preclinical and meta-analytic evidence suggests that lithium may exert protective effects at subtherapeutic concentrations, reinforcing its potential relevance as an environmental exposure of interest for its potential association with reduced dementia risk (De-Paula et al., 2025; Lu et al., 2024). Together, these findings provide both epidemiological and biological support for investigating lithium as a potential modifier of dementia risk at the population level.

Despite growing international evidence, U.S.-based, county-level analyses remain limited, particularly in states with heterogeneous geology such as Texas. Existing studies often rely on national averages, coarse spatial resolution, mortality outcomes, or limited water sampling coverage. Moreover, few studies integrate large-scale environmental monitoring data with public health prevalence data using reproducible statistical modeling frameworks. As a result, important geographic heterogeneity and population-level disease burden may be obscured, limiting the ability to translate findings into targeted public health or environmental policy interventions.

This research aims to integrate the Texas Water Development Board (TWDB) groundwater lithium measurements with county-level dementia prevalence data to develop a regression-based modeling framework. By leveraging a previously constructed, normalized TWDB water-quality database, this study seeks to identify population-level associations between naturally occurring lithium exposure and dementia prevalence across Texas counties, while controlling for demographic and socioeconomic confounders. This work provides a scalable foundation for future environmental health investigations and contributes to ongoing discussions regarding environmental determinants of neurodegenerative disease.

2 Literature Review

2.1 From Psychiatric Treatment to Neuroprotection

Although lithium used at therapeutic levels for bipolar disorder can cause considerable side effects including thyroid dysfunction, renal impairment, and tremor, emerging evidence suggests that much lower doses may confer broader benefits across cardiovascular, metabolic, inflammatory, and cognitive systems (Hamstra et al., 2023). Interest in lithium's potential role in dementia prevention began when psychiatrists

treating elderly patients with bipolar disorder observed that those maintained on lithium for decades had substantially lower rates of ADRD than elderly patients with bipolar disorder who had never received lithium treatment (Nunes et al., 2007). This clinical observation motivated a body of experimental, clinical, and epidemiological research that has grown substantially over the past two decades.

2.2 Experimental Evidence

In an animal study, Andrade Nunes et al. (2013) administered very small daily doses of lithium to mice, including mice genetically predisposed to develop Alzheimer's-like pathology. Compared to untreated mice, those receiving lithium showed less accumulation of abnormal brain protein deposits and reduced brain cell damage in memory-related regions. The authors concluded that long-term microdose lithium may help prevent or slow Alzheimer's-related changes in this model.

Between 2009 and 2012, Forlenza and his team selected 61 physically healthy, older adults at high risk for ADRD, for a randomized study of lithium versus placebo for 2 years (double-blind phase) and followed up for an additional 24 months (single-blinded phase). Participants in the placebo group displayed cognitive and functional decline, whereas lithium-treated patients remained stable after 24 months (Forlenza et al., 2019). More than a decade later, Forlenza and his team (Damiano et al., 2023) followed up, finding 36 of the 61 patients. Even 11 to 15 years after treatment, the patients who had received lithium showed substantially better global cognitive outcomes than those who had received placebo, with large effect sizes. The fact that these differences persisted for more than a decade after the intervention strongly suggests a durable disease-modifying effect, not temporary symptom management. This kind of long-horizon signal is rare in ADRD research and directly challenges the idea that lithium's benefits are modest, transient, or clinically insignificant.

2.3 Clinical Research

Terao et al. (2024) compared the efficacy of low-dose lithium to three recently developed Alzheimer's medications and found that lithium was likely the safest of the four options, further motivating investigation into its possible neuroprotective role.

2.4 Lithium in Drinking Water

Several population-level studies have examined whether long-term exposure to trace lithium in drinking water is associated with differences in dementia rates across populations. Results have been mixed, and meaningful comparisons across studies are complicated by the absence of a shared definition of "high" or "low" lithium exposure. Some studies find an inverse relationship between lithium and ADRD prevalence, while others find non-linear associations with an underlying pattern that remains unclear.

Fraiha-Pegado (2021), in a review of 5 studies of trace levels of lithium in water, noted that concentrations of lithium in drinking water as low as 0.002 mg/L and 0.056 mg/L were associated with a lower risk of mortality from dementia.

Some studies found no relationship. Parker et al. (2018) found no association between concentrations of lithium in groundwater by county and rates of dementia by county. However, a county is an administrative boundary, not a hydrologic unit. Lithium concentrations can vary substantially across aquifers, wells, and municipal systems within the same county. If Parker assigned a single county-mean lithium value to all residents, that assumes uniform exposure within the county, which is rarely true. This kind of spatial averaging can dilute true associations and bias estimates toward the null.

Duthie et al. (2023) studied compared concentrations of lithium in groundwater across Scotland and found no association with dementia. However, the mean lithium level for sampling sites matched to participant data was only 1.26 µg/L (SD 0.63, range 0.55–9.19). Even the upper end of Duthie's range, 9.19 µg/L, is low compared to the mean lithium level for counties in the present study. There is no standardized definition of high or low lithium exposure across studies, and wide variation in exposure ranges limits meaningful comparison of findings.

Some studies found a nonlinear relationship. Kessing et al. (2017) reported that long-term exposure to lithium in drinking water may be associated with a lower incidence of dementia in a nonlinear way. Most recently, Hwang et al. (2026) conducted a nationwide U.S. county-level ecological study using Fifth Unregulated Contaminant Monitoring Rule (UCMR 5) lithium monitoring data and the same NORC dementia prevalence dataset used in the present study and found nonlinear dose-response patterns across exposure levels.

This mixed evidence suggests that lithium's association with dementia outcomes may depend on exposure level, duration, and contextual factors related to water chemistry or geography. One hypothesis is that lithium does not act in isolation, but rather within a broader mineral profile that may modify or interact with its effects. Texas provides an opportunity to examine this question because it offers extensive county-level groundwater mineral measurements, enabling comparison of lithium and co-occurring elements across 254 counties.

The present study seeks to investigate whether co-occurring components of drinking water, such as calcium, magnesium, silica, arsenic, aluminum, and lead, may be modifying lithium's apparent protective association, either enhancing or attenuating its effect on ADRD outcomes. This study examines whether Texas counties with higher groundwater lithium levels have lower dementia prevalence, after accounting for demographic and socioeconomic differences. It also explores whether other water components may modify this association.

3 Methods

3.1 Data Sources

This study uses a cross-sectional observational design to examine the association between naturally occurring groundwater lithium concentrations and county-level dementia prevalence across Texas in 2020. The analysis integrates environmental exposure data from the Texas Water Development Board (TWDB) with Medicare-derived prevalence estimates from the NORC Dementia DataHub.

3.1.1 TWDB Groundwater Quality Database

Groundwater quality data were obtained from the TWDB Groundwater Database, which contains laboratory-measured mineral concentrations from wells across all 254 Texas counties (Texas Water Development Board, 2024). After cleaning, the dataset contained 460,137 rows spanning 1919–2019, with 243 counties represented. Samples collected on or after January 1, 2020, were excluded so that all exposure estimates precede the outcome measurement. Thirteen dissolved minerals were selected based on neurological relevance (lithium, lead, arsenic, manganese), geochemical co-occurrence with lithium (sodium, calcium, magnesium, hardness), and potential modifying roles (fluoride, selenium, aluminum, zinc, potassium). Lithium ($\mu\text{g/L}$, dissolved) is the primary exposure variable.

3.1.2 NORC Dementia DataHub - 2020 County Prevalence

County-level dementia prevalence for 2020 was obtained from the NORC Dementia DataHub, derived from Medicare administrative claims (NORC, 2024). Analysis was restricted to unstratified rows representing the total beneficiary population, classified as highly likely dementia ($\text{DEMENTIACAT} = 1$), among beneficiaries under age 75 ($\text{AGECAT6} \leq 8$). County prevalence was computed as the ratio of prevalent dementia cases to total Medicare beneficiaries. The dataset covered 252 of 254 Texas counties (99.2% coverage).

3.2 Data Preparation

All datasets were harmonized to the county level using Federal Information Processing Standards (FIPS) codes. Counties with missing values in either lithium exposure or dementia prevalence were examined to assess patterns of missingness. Embedded reference rows in the TWDB CSV files were identified and removed. Measurement dates were parsed, and records after December 31, 2019, were excluded to enforce temporal precedence of exposure over outcome. Dementia prevalence data were merged with groundwater exposure data by county FIPS code. Counties missing

either lithium or dementia data were excluded from regression models but retained in descriptive analyses. Prior studies adjusted for demographic and health care resource variables; however, adjustment strategies differed substantially across studies.

3.3 Feature Engineering

For each county and groundwater parameter, summary features were computed over the 2010–2019 exposure window: mean and median concentration, standard deviation, sample count, proportion of samples above the statewide 90th percentile, and a linear trend slope (for counties with ≥ 3 distinct sampling years). This ten-year window captures chronic pre-outcome exposure while reflecting relatively consistent infrastructure and monitoring practices.

3.4 Exposure Homogeneity Classification

A key methodological concern in county-level environmental studies is exposure misclassification: county-mean lithium averages residents drawing from wells with very different concentrations, attenuating regression coefficients toward zero even when a true biological effect exists. This is a form of classical measurement error in the predictor variable. To address this, counties were classified as having homogeneous groundwater lithium if they met all the following criteria, applied to the 2000–2019 measurement window to maximize sample size for classification:

- At least 10 lithium samples collected during 2000–2019
- At least 70% of wells in the HIGH category (≥ 65.00 $\mu\text{g/L}$) or the LOW category (≤ 6.80 $\mu\text{g/L}$)
- Coefficient of variation ($\text{CV} = \text{SD} / \text{mean}$) ≤ 0.50
 - Environmental threshold for “low dispersion.”

Table 1. Homogeneous Lithium Exposure Counties (2000–2019). Counties meeting all criteria were designated HIGH or LOW lithium. The statewide 25th and 75th percentile thresholds for lithium concentration were 6.80 $\mu\text{g/L}$ and 65.00 $\mu\text{g/L}$, respectively. Fourteen counties qualified: 12 HIGH and 2 LOW (Kinney and Real). Table 1 lists these counties with their group assignments.

<i>County</i>	<i>Group</i>	<i>Mean Li ($\mu\text{g/L}$)</i>	<i>CV</i>	<i>N Samples</i>
<i>Bailey</i>	HIGH	126.3	0.48	35
<i>Briscoe</i>	HIGH	104.4	0.37	14
<i>Dallas</i>	HIGH	73.3	0.29	21

<i>Floyd</i>	HIGH	97.9	0.37	39
<i>Hale</i>	HIGH	79.9	0.21	39
<i>Hardeman</i>	HIGH	82.4	0.31	10
<i>Hockley</i>	HIGH	134.5	0.33	39
<i>Kenedy</i>	HIGH	69.2	0.10	10
<i>Kinney</i>	LOW	2.4	0.25	22
<i>Lubbock</i>	HIGH	130.5	0.49	34
<i>Ochiltree</i>	HIGH	138.2	0.42	28
<i>Real</i>	LOW	2.2	0.14	12
<i>Starr</i>	HIGH	125.4	0.40	22
<i>Swisher</i>	HIGH	90.0	0.38	35

Note: Statewide thresholds: LOW ≤ 6.80 $\mu\text{g/L}$ (p25); HIGH ≥ 65.00 $\mu\text{g/L}$ (p75). CV = coefficient of variation.

3.5 Exploratory Data Analysis

Exploratory analyses characterized the distribution and geographic patterns of both primary variables and examined pairwise correlations among all 13 groundwater parameters. Analyses included distributional assessments, identification of outliers, county-level spatial visualizations, and preliminary correlation analysis between lithium exposure and dementia prevalence. Choropleth maps were produced for lithium concentration and dementia prevalence using 2023 Census TIGER/Line county boundaries (Figure 2). Visual inspection reveals substantially higher lithium concentrations in the Texas Panhandle and West Texas regions, with lower concentrations in central and eastern counties. A mineral correlation matrix (Figure 1) assessed multicollinearity relevant to model specification. The most notable relationships include a near-perfect correlation between hardness and calcium ($r \approx 0.98$), a strong magnesium–hardness association ($r \approx 0.91$), and moderate lithium–sodium ($r = 0.56$) and lithium–potassium ($r = 0.68$) correlations. These patterns directly informed covariate selection and the decision to exclude calcium and hardness from the primary model specification.

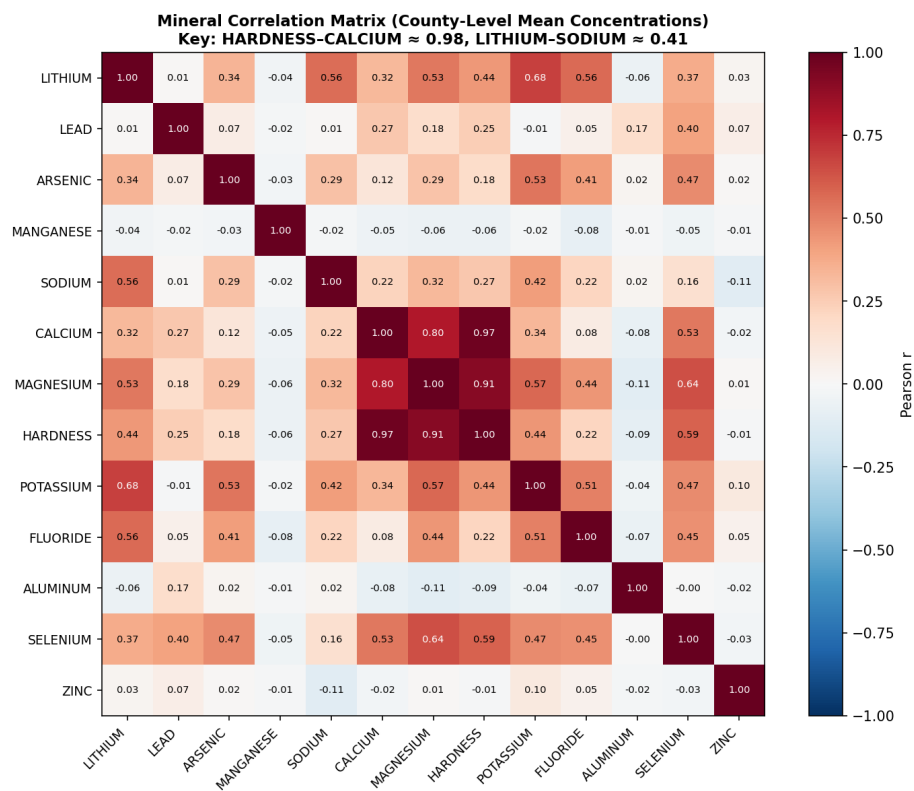


Fig. 1. Pearson correlation matrix of county-level mean concentrations for 13 groundwater minerals across Texas counties (TWDB, pre-2020). Color intensity reflects the strength of correlation (red = positive, blue = negative). Notable relationships include hardness–calcium ($r \approx 0.98$), magnesium–hardness ($r \approx 0.91$), and lithium–sodium ($r = 0.56$), informing covariate selection in regression models.

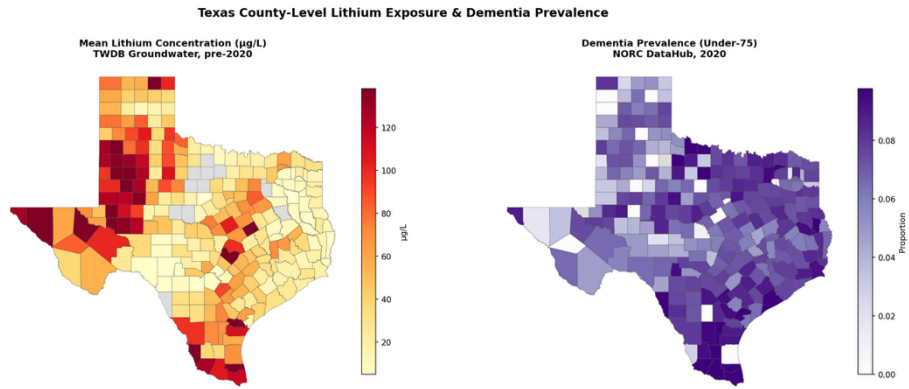


Fig. 2. Side-by-side choropleth maps of Texas counties displaying mean groundwater lithium concentration ($\mu\text{g/L}$, pre-2020; TWDB, 2024) and 2020 dementia prevalence among Medicare beneficiaries under age 75 (NORC, 2024). Gray counties indicate missing lithium data. Higher lithium concentrations are concentrated in the Panhandle and Trans-Pecos regions.

3.6 Regression Modeling

Ordinary least squares (OLS) regression with heteroskedasticity-consistent standard errors was used to assess the lithium–dementia association. Three nested model specifications were fit on both the full county sample ($N = 240$) and the homogeneous subsample ($N = 14$):

- Model A: dementia prevalence $\sim$ lithium
- Model B: dementia prevalence $\sim$ lithium + sodium (primary specification)
- Model C: dementia prevalence $\sim$ lithium + sodium + calcium + magnesium + hardness

Full-sample models are expected to produce attenuated lithium coefficients due to exposure misclassification. Homogeneous subsample models reduce this attenuation by restricting to counties where the county mean is a more valid exposure proxy for the population. Sodium is included in Model B because it shares lithium’s physiological transport pathways and is moderately correlated with lithium ($r = 0.56$), making it both a potential confounder and a geochemical co-variate. Model C is included as a sensitivity check but is susceptible to collinearity instability given the near-perfect hardness–calcium correlation ($r \approx 0.98$) and the small subsample size.

3.7 Nonparametric Inference

Because the homogeneous subsample contains only 14 counties, asymptotic p-values are unreliable. Two nonparametric procedures supplement the OLS results: (1) a one-tailed permutation test (10,000 permutations) assessing whether the observed lithium slope is more negative than expected under the null, and (2) a bootstrap 95% confidence interval for the lithium coefficient (10,000 resamples with replacement). The proportion of bootstrap slopes in the negative (protective) direction is reported as a directional support metric.

3.8 Influence Diagnostics

Given the small homogeneous subsample, Cook's distance and leave-one-out (LOO) analysis was conducted under Model B to assess whether any single county drives the direction or magnitude of the lithium coefficient. A Cook's distance threshold of $4/N \approx 0.286$ was used. The proportion of LOO slopes remaining in the protective direction is reported as a robustness check.

3.9 Software

All analyses were conducted in Python using pandas, NumPy, statsmodels, scipy, matplotlib, and geopandas. Census data were accessed via the U.S. Census Bureau API. All code is maintained in the shared EDA notebook (Water_Quality_Dementia_Analysis.ipynb) and is designed to be fully reproducible from the raw TWDB, NORC, and Census data sources.

4 Results

4.1 Descriptive Statistics

After data preparation, the analytic dataset contained dementia prevalence estimates for 252 Texas counties (99.2% coverage) and lithium data for 242 counties (95.3% coverage). The full regression sample included 240 counties with both measures available. Across all counties, the median under-75 dementia prevalence was 7.19%, with a range of 0.00% (Borden County, very small Medicare population) to 16.65% (Starr County). The median county-mean groundwater lithium concentration was 38.31 µg/L, reflecting substantial geographic variation driven by differences in underlying geology and aquifer systems across Texas.

4.2 Full-Sample Regression Results

Across 240 Texas counties with complete data, mean groundwater lithium concentration showed a weak negative correlation with dementia prevalence (Pearson $r = -0.095$), consistent with a mild inverse association. However, OLS regression results in the full sample showed minimal explanatory power across all three model specifications (Table 2), as expected under exposure misclassification. Model A (lithium only) yielded a lithium coefficient of -0.000042 ($p = 0.21$, $R^2 = 0.009$). In Model B (lithium + sodium), the lithium coefficient increased in magnitude to -0.000123 ($p = 0.056$, $R^2 = 0.082$), with sodium absorbing substantial variance in dementia prevalence.

4.3 Homogeneous Subsample Regression Results

In the homogeneous-county subsample ($N = 14$), Model B, yielded a lithium coefficient of -0.000112 (HC3 SE = 0.000317, $p = 0.73$, $R^2 = 0.226$). Although not statistically significant by conventional thresholds, the coefficient is directionally consistent with a protective association. Model A in the homogeneous subsample showed a sign reversal (positive lithium coefficient), reflecting the confounding influence of sodium—sodium is geochemically co-distributed with lithium and is also associated with higher dementia prevalence in this small sample, creating an upward confounding effect that is resolved when sodium is included. Model C was estimated but is not interpreted due to severe collinearity among calcium, magnesium, and hardness at $N = 14$.

Table 2. Model Comparison Summary. *Blue shading = full sample; orange shading = homogeneous subsample. Homogeneous counties: Bailey, Briscoe, Dallas, Floyd, Hale, Hardeman, Hockley, Kenedy, Kinney, Lubbock, Ochiltrie, Real, Starr, Swisher.*

<i>Model</i>	<i>N</i>	<i>Li Coef</i>	<i>SE (HC3)</i>	<i>p-value</i>	<i>R²</i>	<i>Adj R²</i>
<i>A — Full: dementia ~ Li</i>	240	−0.000042	0.000034	0.2092	0.0090	0.0048
<i>B — Full: dementia ~ Li + Na</i>	240	−0.000123	0.000065	0.0563	0.0823	0.0746
<i>C — Full: dementia ~ Li + Na + Ca + Mg + Hard</i>	240	−0.000076	0.000073	0.2988	0.1028	0.0861
<i>A — Homog: dementia ~ Li</i>	14	+0.000089	0.000332	0.8123	0.0087	−0.074
<i>B — Homog: dementia ~ Li + Na</i>	14	−0.000112	0.000317	0.7296	0.2257	0.0850
<i>C — Homog: dementia ~ Li + Na + Ca + Mg + Hard</i>	14	See note	(collinear)	N/A	N/A	N/A

4.4 Nonparametric Inference

Given the small subsample size, nonparametric procedures were used to assess the robustness of the lithium coefficient direction. The one-tailed permutation test (10,000 permutations) yielded $p = 0.3493$, indicating that slopes as negative as the observed value occur in approximately 35% of random permutations—not statistically unusual, but also not inconsistent with a weak protective signal given the extreme power constraints at $N = 14$. The bootstrap 95% confidence interval for the lithium coefficient was $[-0.00065, +0.00055]$, spanning zero and confirming the uncertainty. Notably, 65.6% of the 10,000 bootstrap resampled slopes were negative (protective), indicating a directional majority consistent with the hypothesized association. Leave-one-out analysis found that 86% of LOO slopes (12 of 14 models with each county removed in turn) remained in the protective direction, suggesting the negative direction is not driven by any single influential county.

5 Discussion

5.1 Interpretation of Results

The results of this study are directionally consistent with the hypothesis that naturally occurring groundwater lithium is inversely associated with dementia prevalence, but statistical evidence is not conclusive at the county level. The full-sample null result, a near-zero lithium coefficient across 240 Texas counties, does not contradict the prior literature and is in fact expected. County-mean lithium is a poor proxy for individual-level chronic exposure when wells within a county vary widely; classical measurement error in the predictor attenuates regression coefficients toward zero regardless of the true biological effect size.

In the homogeneous subsample ($N = 14$), the primary model (Model B: dementia $\sim$ lithium + sodium) yielded a negative lithium coefficient, with 65.6% of bootstrap resampled slopes and 86% of leave-one-out slopes also in the protective direction. This directional consistency is more meaningful than the p-value alone given the extreme power constraints of an $N = 14$ ecological analyses. The sign reversal in Model A, where the lithium coefficient becomes positive before sodium is controlled for, illustrates the importance of adjusting for sodium, which is geochemically co-distributed with lithium and independently associated with dementia prevalence in this sample.

One of the more striking descriptive findings is the geographic patterning visible in Figure 2: counties in the Texas Panhandle and West Texas, sitting atop geologically old, lithium-rich aquifer systems, tend to have lower dementia prevalence than counties in Central and East Texas. This spatial pattern is suggestive but requires formal spatial regression analysis to disentangle from correlated geographic factors such as population age structure, rurality, and healthcare access.

5.2 Implications

These findings have implications at several levels. For environmental health surveillance, they suggest that passive groundwater monitoring data already collected by the TWDB contains signal relevant to neurodegenerative disease outcomes – a connection that has not previously been systematically explored in Texas. For public health planning, the geographic clustering of low dementia prevalence in the Panhandle and West Texas, regions overlying lithium-rich aquifers, raises the possibility that naturally occurring mineral exposure is one contributor to spatial heterogeneity in ADRD burden. While this study does not establish causation, it provides geographically grounded motivation for further investigation in a state with both the data infrastructure and the geological diversity to advance this line of research.

Perhaps the most unexpected finding is the magnitude of the R^2 increase when moving from the full county sample to the homogeneous subsample: from 0.009 (Model A, full sample) to 0.226 (Model B, homogeneous subsample). This jump, achieved simply by restricting to counties where the county mean is a more valid proxy for individual exposure, illustrates how severely exposure misclassification can mask a true biological signal in ecological studies. The sign reversal in Model A of the homogeneous subsample, where the uncontrolled lithium coefficient is positive, is equally instructive: it demonstrates that sodium confounding alone is sufficient to reverse the apparent direction of the lithium–dementia association, and that controlling for geochemical co-variables is not optional but essential in this analysis. These methodological lessons extend beyond lithium and may apply broadly to county-level studies of trace elements and health outcomes.

5.3 Limitations

5.3.1 Exposure Pathways and Spatial Misclassification in Ecological Lithium Studies

Lithium levels in groundwater may not always correlate 1:1 with lithium in tap water. The consumption of tap water varies by community, socioeconomic status, and decade. Individuals may not consume all or even most of their drinking water in the area where they sleep; they may travel to work or school for much of the day and drink water from another region. Aquifers used for one population now may have been used for a much larger geographic distribution decades before.

5.3.2 Dietary Lithium Sources Beyond Groundwater

Groundwater is one source of dietary lithium, though most intake comes from food (Greenblatt, 2025). The amount of lithium in groundwater is correlated with the amount of lithium in foods grown in that area; however, the proportion of one's diet that comes from local foods varies considerably by socioeconomic class and decade with all groups moving toward increasingly globally diverse diets in recent years.

5.3.3 Residential Mobility and Exposure History

The ADRD prevalence data used in the present study ties a person diagnosed with dementia to a county by their residence at the time of diagnosis. From this anonymized data, it is not possible to determine how long the individual lived in that county before the diagnosis. It is likely that for many individuals, they lived in a high lithium area for decades and then moved to a low lithium area or vice versa, sometime before the prevalence data was collected.

5.3.4 Diagnosing a human being with a disease is like painting a Venn diagram on moving water.

As ADRD is rare before the sixth decade, very young patients presenting with dementia symptoms will likely be sent to a specialty clinic and undergo extensive neurological evaluation before being diagnosed. A very elderly patient may not get any sort of testing beyond a visit to their family practice doctor. Additionally, diagnostic criteria vary by region, by medical specialty, and by the availability of specialists and specialized tests. The availability of specialists varies by region, and the accessibility of those specialists and their specialized test varies by socioeconomic class.

Present in the literature are detailed explorations of the possible biological mechanisms by which lithium reduces the risk of Alzheimer's Disease specifically. Among the other dementias, there are other mechanisms. Dementia caused by vascular disease or Lewy bodies or prions may be less impacted by lithium levels than dementia caused by Alzheimer's Disease.

5.3.5 Exposure Misclassification due to Spatial Aggregation

Counties are lines humans draw on maps without regard to the distribution of lithium in groundwater. To protect patient privacy, the most readily available data for the prevalence of ADRD is anonymized to the county level. This makes it difficult to compare levels of lithium to ADRD prevalence at a tighter spatial grouping.

5.4 Future Research

The most important methodological advance needed for this line of research is disaggregation: we must move beyond county-level prevalence data and county-level lithium data. Smaller spatial units, such as, zip codes, census tracts, or water service area boundaries, can more closely match actual exposure zones. Linking individual Medicare or Medicaid claims to the specific public water systems serving each beneficiary's address, combined with UCMR 5 lithium monitoring data, would allow far more precise estimation of the lithium–dementia association at the individual level. Future studies should also incorporate neurologist density as a proxy for diagnostic capacity (Lepping et al., 2015), which Hwang et al. (2026) included and which the present study does not yet account for, though this data is available for future iterations.

From a policy perspective, the question of whether the EPA should establish a health advisory or drinking water standard for lithium deserves serious consideration as the evidence base grows. Establishing routine monitoring of lithium in water supplies—analogueous to current monitoring of iron, fluoride, and nitrate—would provide the longitudinal exposure data needed to conduct stronger epidemiological analyses. Further out, questions about dietary reference targets and whether fortification

of the food supply with trace lithium, as is done with iodine, iron, thiamine, and vitamin D, will require a much larger and more precisely designed evidence base than currently exists.

5.5 Ethical Considerations

This study raises ethical considerations relevant to environmental health research more broadly. The decision to examine whether a naturally occurring substance in public water supplies may reduce the burden of a devastating neurodegenerative disease carries both scientific and normative dimensions. First, ecological research designs like this cannot establish causation; reporting directional associations without appropriate qualification risks being misinterpreted by the public or policymakers in ways that outpace the evidence. We have attempted to be explicit throughout about the limitations of this design and the distinction between directional signal and statistical certainty.

Second, any future policy discussion about lithium in water supplies must be grounded in a careful weighing of potential benefits and risks. Lithium at therapeutic doses carries well-documented side effects; the assumption that trace environmental exposure is categorically safe warrants investigation, not assumption. Conversely, a reflexive precautionary framing that treats any unmeasured exposure as a risk could impede research into a potentially protective factor. The ethical path forward is rigorous, well-funded, transparent research and not premature intervention in either direction.

Third, there are equity dimensions to this research. The counties identified as having homogeneous high-lithium groundwater in this study, largely in the Panhandle and West Texas, are predominantly rural and lower-income, with limited access to specialty care. If protective mineral exposure is concentrated in areas that are otherwise underserved, that geographic coincidence does not constitute equitable access to prevention. Any translational benefit from this line of research must be designed with geographic and socioeconomic equity as an explicit goal, not an afterthought.

6 Conclusion

This study presents early evidence that county-level groundwater lithium concentration is directionally associated with lower dementia prevalence in Texas, with the most informative results coming from the 14 counties classified as having homogeneous groundwater lithium exposure. In this subsample, 65.6% of bootstrap resampled regression slopes and 86% of leave-one-out slopes were in the protective (negative) direction, consistent with the broader epidemiological and experimental literature on lithium's potential neuroprotective properties. However, the statistical uncertainty is substantial at this spatial scale, and no conclusions about causation can be drawn from this ecological design.

The primary methodological contribution of this work is the exposure homogeneity classification framework, which provides a principled approach to reducing exposure misclassification in county-level ecological studies of environmental trace elements. This framework is extensible to other minerals and other health outcomes. Future research should build on this foundation by incorporating finer spatial resolution, longitudinal exposure data, and individual-level linkage to clinical records. From a policy perspective, the findings support continued monitoring of lithium in public water supplies and motivate investment in larger, more precisely designed epidemiological studies before any intervention recommendations can be made.

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